A comparative analysis of the USRP N210 pilot dataset against published WiFi sensing benchmarks

CSI Human Activity Recognition project

July 20, 2026

The question, and the short answer

What we asked

We captured 8 CSI takes on a USRP N210 and got 98–99% accuracy. Is the *data* actually sound, or did we get a good number for the wrong reasons?

What holds up

- Capture rate is sufficient for the physics of this task — proven, not assumed (slide 9)
- Motion signal is strong and real: 4.2× energy separation
- Evaluation protocol is stricter than ARIL's and UT-HAR's
- Front-end choices match a published USRP dataset (ARIL)

What does not

- Task is nearly trivial: **one scalar gets 99.1%**. Our 98–99% proves the pipeline, not the model
- 13% of windows silently discarded by a preprocessing misconfiguration
- 4×–800× smaller than reference datasets
- 2 subjects, 1 room, 1 day ⇒ no domain diversity

Verdict

The collection method is sound and the pilot passes. The accuracy is real but *uninformative* — it validates the chain, not the approach. The 7-class plan needs the slide-16 fixes.

What we collected

Setup. USRP N210 *passively sniffs* an existing MacBook → Android-hotspot 802.11 link. Channel 11 (2.462 GHz), 20 MHz, single antenna. 52 OFDM subcarriers, amplitude only.

Task. Binary: `action` (person moving) vs `noaction` (person present but still). Presence is held constant on purpose — the model must learn *motion*, not *occupancy*.

Takes. 2 subjects × 4 takes × ≈90 s: `walk`, `mixed` (squat → bend → drink → side-steps, ≈18–20 s loop), `stand`, `sit`.

Take	Subject	Hz	s
walk	rishiikesh	77.6	91.6
mixed	rishiikesh	56.1	95.9
walk	amrith	73.0	91.9
mixed	amrith	73.5	107.9
stand	rishiikesh	62.9	89.8
sit	rishiikesh	51.5	90.9
stand	amrith	65.6	92.1
sit	amrith	71.0	91.8
total		**66.4**	**752**

red = action, blue = noaction. 322 windows of 2 s survived segmentation.

Data hygiene check

Every file carries **exactly one truncated final row** (Ctrl+C landed mid-write) — 361 empty cells out of 3.1 M, i.e. 0.015%. No zero rows, no constant or dead subcarriers. **Cosmetic, not a data-quality problem**; drop the last row.

Which datasets we compare against, and why

Three published datasets that bracket our setup from different directions. All quoted figures are the authors' published specifications.

ARIL (2019) — the hardware twin

Ettus **USRP N210**, **52 subcarriers**, **single TX/RX link**, **amplitude only**, segments linearly interpolated to fixed length. *Almost every low-level choice matches ours.* Difference: they *control* both ends (two USRPs + OctoClock GPS sync); we sniff a live link passively. 6 hand gestures, 16 locations, **1 volunteer**, 1394 samples.

UT-HAR / Yousefi et al. (2017) — the task twin

Intel 5300 NIC, $30 \times 3 = 90$ streams, **1 kHz**. Seven whole-body activities (walk, run, fall, sit down, stand up, lie down, pick up) — the closest published task to our eventual 7-class plan. 6 subjects, ≈4,477 samples.

Widar 3.0 (2019) — the gold standard

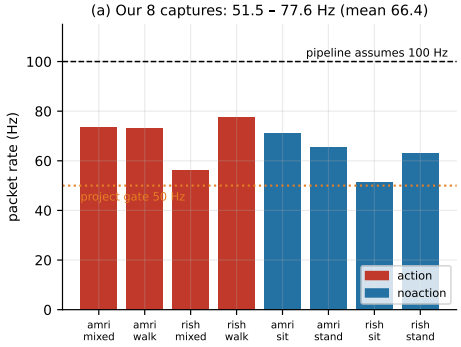
Intel 5300, 1 TX + 6 RX × 3 antennas = **18 links** × 30 subcarriers, **1 kHz**. 22 gestures, **16 subjects, 75 domains**, **258k instances**. Uses *phase* across antennas for Doppler (DFS) and velocity profiles (BVP) — which our single-antenna setup structurally cannot reproduce.

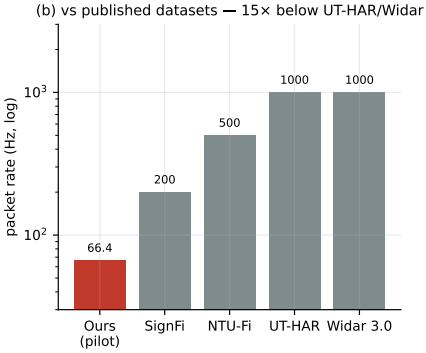
Head-to-head specification table

	Ours (pilot)	**ARIL**	**UT-HAR**	**Widar 3.0**
Radio	USRP N210	USRP N210 ×2	Intel 5300	Intel 5300
Acquisition	**passive sniff**	controlled TX+RX	controlled	controlled
Sync / clock	none	OctoClock GPS	NIC	NIC
Subcarriers	52	52	30	30
Links × antennas	1 (SISO)	1 (SISO)	3	18
Total CSI streams	52	52	90	540
Packet rate	51–78 Hz	not reported	1000 Hz	1000 Hz
Nyquist (worst)	25.7 Hz	—	500 Hz	500 Hz
Uses phase	× (unsynced)	× (discarded)	×	✓ DFS/BVP
Classes	2	6	7	22
Subjects	2	1	6	16
Rooms / domains	1	16 positions	1	75
Days	1	n/r	n/r	multiple
Labelled samples	322	1,394	4,477	258,000
Test split	**leave-one-subject-out**	every 5th sample	random	cross-domain

Read this honestly: we lose badly on *scale* and *diversity*, we match ARIL on *front-end design*, and we are the *only one of the four* whose headline number comes from a subject-independent split.

Capture rate — the headline weakness





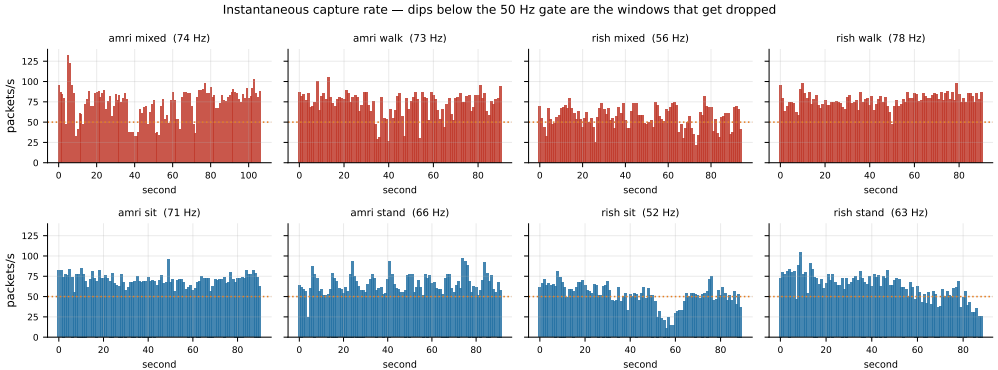
Measured: 51.5–77.6 Hz, mean **66.4 Hz**. All 8 takes clear the project's own 50 Hz gate, but every take is *below* the 100 Hz the preprocessing pipeline assumes.

Cause: we sniff a real link opportunistically. Rate is set by how many frames the MacBook emits *and* that `gr-ieee802-11` can decode (legacy OFDM only; 802.11n HT frames are dropped). Widar/UT-HAR simply command a 1 kHz injector.

But is 66 Hz actually too slow? — see slide 9 before concluding

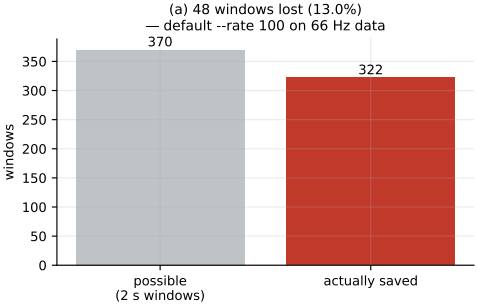
It is 15× below UT-HAR. That sounds fatal. Measured against the *physics of what we are sensing*, it turns out not to be — for *this* task.

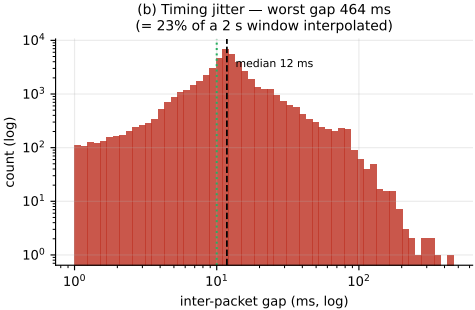
Rate stability — where the dropped windows came from



Rate is *not* steady. Per-second minima fall to **11–55 packets/s** (`rishiikesh sit` bottoms out at 11). Every dip below the segmenter's fill threshold silently deletes a 2 s window. **This is a capture-side artefact of passive sniffing**, not a bug — but it must be budgeted for, not ignored.

Timing jitter and the 13% we threw away





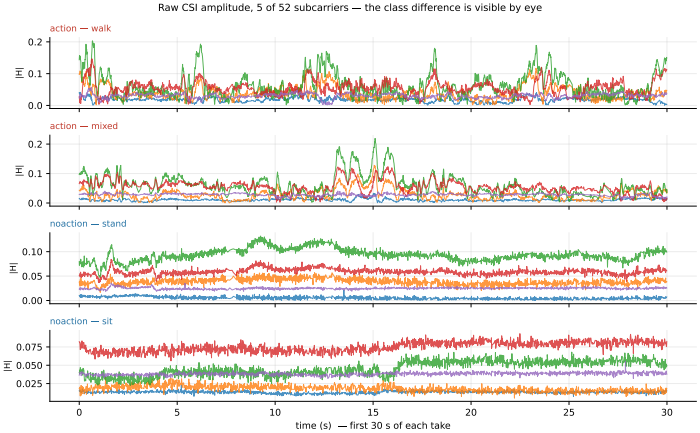
A real, fixable mistake. `segment_all_pilot.py` ran `segment_windows.py` at its *default* `--rate 100`. On 66 Hz data each 2 s window then needs $\geq$100 packets to survive; many did not.

752 s of capture $\Rightarrow$ 370 possible windows, **322 saved, 48 lost (13.0%)**.

Fix: pass `--rate 50`. Threshold drops to 50 packets/2 s, which every take clears. Recovers $\approx$48 windows (+15% data) for free, with no recapture.

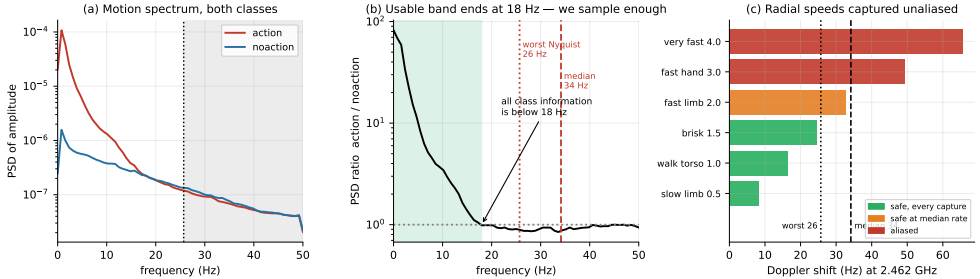
Jitter: median gap 11–14 ms, but the 99th percentile reaches 99 ms and the worst gap is 464 ms — 23% of a 2 s window filled by interpolation across a hole.

The signal itself is real



The project's own "wiggle test" criterion — *if you cannot see it by eye, the network cannot either* — is comfortably met. `walk` and `mixed` show large multi-scale fluctuation; `stand` and `sit` are visibly flat (residual ripple is breathing plus noise).

Is 66 Hz enough? Measuring, not guessing

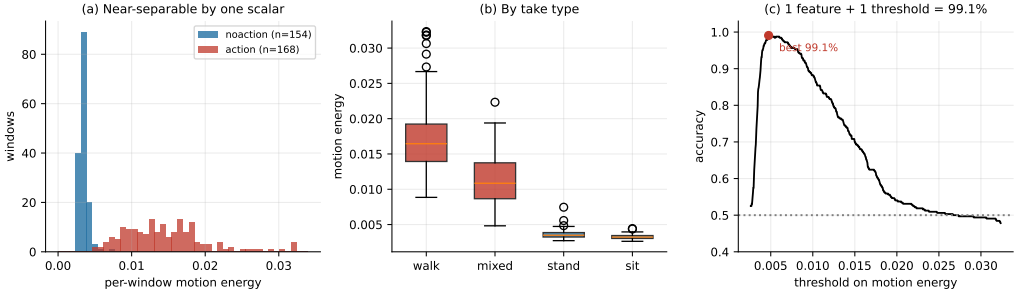


The key result of this analysis

Panel (b) divides the `action` spectrum by the `noaction` spectrum. Above **18 Hz the ratio is 1.0** — the classes are *indistinguishable*; that band is pure noise floor. **All class-discriminative information lives below 18 Hz.** Our worst-case Nyquist is **25.7 Hz**, median **34 Hz** ⇒ **we sample the informative band with 1.4–1.9× headroom.** The 15× rate deficit versus UT-HAR costs *nothing measurable on this task*.

Caveat for the 7-class plan: this holds because `walk` and `mixed` are gross whole-body motions (Doppler $\lesssim$25 Hz, panel c). A fast hand (3 m/s → 49 Hz) or a `nod` lands *outside* this band and aliases. **We have not yet proven those classes are capturable at 66 Hz.**

How separable are the classes, really?

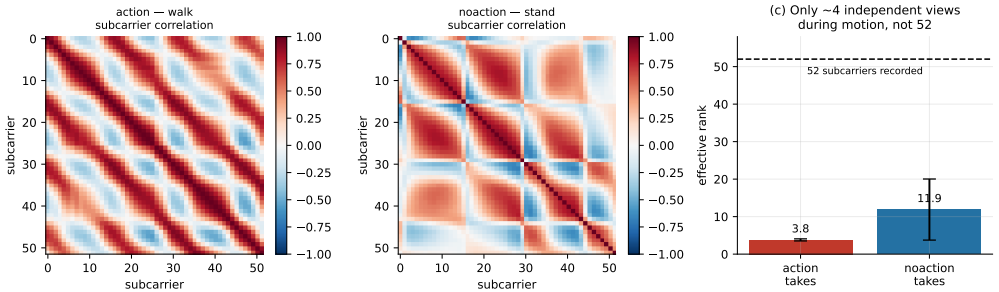


The uncomfortable finding — state this to anyone quoting our accuracy

Take one scalar per window: mean over subcarriers of the temporal standard deviation ("motion energy"). `action` averages 4.20× `noaction` (0.0145 vs 0.0035), Fisher ratio 3.85.
A single threshold on that one number classifies 99.1% of windows correctly.

Our CNN–GRU–Attention model scores 98.4–99.3%. **It does not beat the one-line baseline.** That is not a failure of the model — it means *this binary task cannot discriminate between a good model and a trivial one.* Per-take energies (panel b) show why: `walk` 0.0174 and `mixed` 0.0114 sit far above `stand` 0.0036 and `sit` 0.0033.

52 subcarriers, but how many independent views?



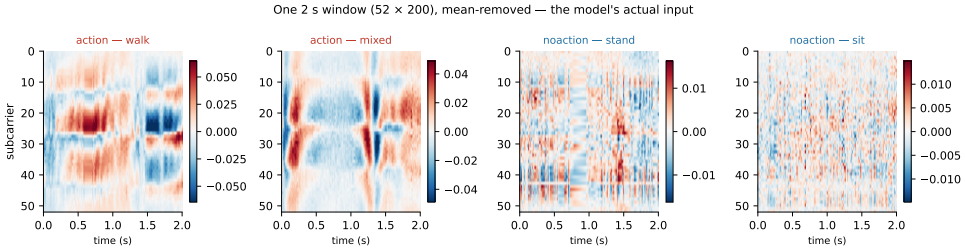
Neighbouring subcarriers are strongly correlated (mean $|r|$ = 0.38–0.45 during motion). Effective rank of the 52-dimensional covariance: **3.5–4.2 during** `action`, 5.6–25.7 during `noaction`.

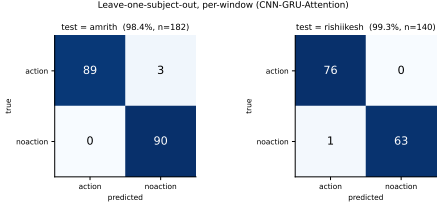
A moving body imposes *one* dominant channel perturbation that all subcarriers see together — so 52 measurements carry roughly **4 independent dimensions**.

Why this matters. The ImgFi recipe multiplies the dataset ×52 by making one recurrence image per subcarrier and calls it free data. It is not free: those 52 images are ≈4 distinct views plus 48 near-duplicates.

The effective dataset is closer to **322 windows** × **4**, not ×52. This is a strong argument for the team's move to the whole-matrix CNN–GRU model, which consumes all 52 jointly instead of voting over them.

What the model actually sees, and what it scored





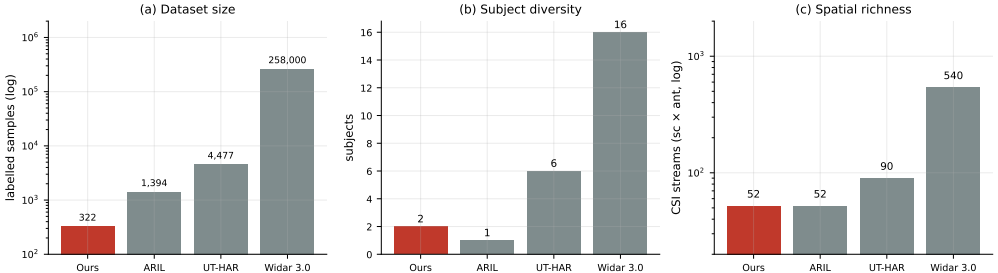
Leave-one-subject-out, per window:

- test = amrith: **98.4%** (179/182)
- test = rishiikesh: **99.3%** (139/140)
- **Headline (worse of the two): 98.4%**

Against the pilot's own pass bar ($\geq$95% $\Rightarrow$ green light) this passes clearly. Only 4 errors in 322 windows, and errors are not concentrated in one class.

Read together with slide 10: a 1-feature baseline gets 99.1% on the same data.

Scale and diversity — where we are genuinely behind



- **Volume.** 322 samples vs 1,394 (ARIL), 4,477 (UT-HAR), 258,000 (Widar 3.0) — 4× to 800× smaller.
- **Subjects.** 2 — exactly one leave-one-subject-out fold each direction, with no third subject to confirm the result generalises.
- **Domains.** One room, one geometry, one orientation, one evening. Widar's entire contribution is the *75 domains* that expose how badly WiFi sensing overfits to environment. We cannot detect that failure mode at all.
- **Streams.** 52 vs Widar's 540. Single-antenna SISO also means no antenna-pair phase difference ⇒ DFS/BVP features are unavailable in principle, not merely unimplemented.

Where our methodology is stronger than the references

Being honest cuts both ways. On evaluation protocol we are the strictest of the four.

Dataset	Reported split	Leakage risk
Ours	leave-one-subject-out	none — subject never seen in training
ARIL	every 5th sample to test	high — same volunteer, same session, adjacent reps
UT-HAR	random train/test	high — overlapping sliding windows can straddle the split
Widar 3.0	cross-domain	low — explicit unseen room/orientation/user protocols

Two design decisions that were correct

1. Amplitude only. Our USRP is not synchronised to the MacBook, so raw phase is scrambled by CFO/SFO/timing offset. We discard it. *ARIL discards phase too — even though they had OctoClock GPS sync and controlled both ends.* Independent confirmation that this was the right call.

2. Person present in both classes. `noaction` is a still *person*, not an empty room. This forces the model to learn motion rather than occupancy — a harder and far more useful problem than the trivial version.

Honest scorecard

Aspect		Assessment
Hardware / front-end	✓	Matches ARIL, a peer-reviewed USRP dataset, spec for spec
Amplitude-only choice	✓	Correct and independently corroborated
Sampling adequacy	✓	Measured 1.4–1.9× headroom over the 18 Hz informative band
Signal strength	✓	4.2× energy separation; visible by eye
Class design	✓	Motion-vs-stillness with presence held constant
Evaluation protocol	✓	Strictest of the four datasets compared
Data hygiene	✓	0.015% truncated cells; otherwise clean
Rate stability	∼	Dips to 11–55 pkt/s; gaps to 464 ms
Window yield	∼	13% lost to a preprocessing default — recoverable without recapture
Subcarrier redundancy	∼	≈4 effective dimensions, not 52
Task difficulty	×	99.1% from one scalar ⇒ accuracy proves nothing about the model
Dataset size	×	4–800× smaller than references
Subject diversity	×	2 subjects; ARIL has 1, but Widar has 16
Domain diversity	×	1 room, 1 day, 1 geometry — overfitting is undetectable
Spatial diversity	×	SISO; no DFS/BVP possible even in principle

What to change before the 7-class collection

Must fix — costs nothing

1. **Segment with** `--rate 50`, not the default 100. Recovers ≈13% of windows immediately.
2. **Always report a trivial baseline** (motion energy + threshold, or a small random forest) beside the deep model. If the gap is not large, the task is too easy to justify the architecture.
3. **Drop the truncated last row** of every capture in the loader.
4. **Fix** `segment_all_pilot.py`**'s glob:** it matches `*.csv`, so the `*_confusion.csv` files now sitting in `data/pilot/` would be parsed as captures and create a bogus `windows/model/` class. Restrict it to `P01_*.csv`.

Must fix — costs collection time

5. **Pre-flight each new class before bulk collection.** Record 30 s of `nod` and `drink` and run the slide-9 spectral-ratio test. If a class has no `action`/`noaction` ratio above 1 inside 0–25 Hz, *it is not capturable at our rate* — find that out in 10 minutes, not after a full session.
6. **Raise the packet rate if that test fails.** Force legacy-OFDM traffic, shorter ping interval, or inject with a second USRP as ARIL did.
7. **$\geq$5 subjects, $\geq$2 days, $\geq$2 room geometries.** Without a second day and a second geometry we cannot measure the failure mode Widar exists to expose.
8. **Use** `--step 1.0` for 50% window overlap — roughly doubles sample count at zero capture cost (train-side only; keep test windows non-overlapping).

Conclusion

Did we collect the data correctly?

Yes. The front-end design, the amplitude-only decision, the class definitions and the evaluation protocol are all sound, and several match or exceed published practice. The one concrete error — segmenting 66 Hz data with a 100 Hz assumption — is fixable without recapturing anything.

Does the 98.4% mean what it looks like?

No. It is an honest, leakage-free number, but a single hand-crafted scalar reaches 99.1% on the same windows. The pilot proves the *chain works end to end*. It does *not* show that our model, our features, or our sampling rate will survive 7 classes.

What the pilot was for

Exactly this. It was designed as a throwaway sanity check, and it did its job: capture, segmentation, training and honest evaluation all work. **Green light for the 7-class collection**, conditional on the slide-16 fixes — above all the 10-minute spectral pre-flight on `nod` and `drink`, which is where this setup is most likely to fail.

References and reproducibility

Reference datasets (all figures quoted are published specifications, not re-measured):

1. F. Wang, J. Feng, Y. Zhao, X. Zhang, S. Zhang, J. Han. *Joint Activity Recognition and Indoor Localization with WiFi Fingerprints* (ARIL). IEEE Access, 2019. arXiv:1904.04964.
2. S. Yousefi et al. *A Survey on Behavior Recognition Using WiFi Channel State Information* (UT-HAR). IEEE Communications Magazine, 2017.
3. Y. Zheng et al. *Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi* (Widar 3.0). ACM MobiSys, 2019.
4. J. Yang et al. *SenseFi: A library and benchmark on deep-learning-empowered WiFi human sensing*. Patterns, 2023. (Used for cross-dataset rate/scale figures.)

Our numbers — every pilot figure in this deck is measured, not quoted:

- `analysis/analyze_pilot.py` → `analysis/pilot_stats.json` (rates, jitter, correlation, effective rank, separability, window yield)
- `analysis/make_figures.py` → `analysis/figures/*.pdf`
- Source data: 8 CSVs and 322 windows under `data/pilot/`
- Model results read directly from `hybrid_model_test-*_confusion.csv`

Limitation of this analysis: reference datasets are compared on published specifications. We did not download and re-analyse their raw CSI, so their signal-level statistics (jitter, effective rank, spectral ratio) are not measured here.